

Project 1:

Student Performance Analysis Using Python

1. Introduction

This project focuses on analyzing student academic performance using data analysis and visualization techniques in Python. The dataset includes student scores in mathematics, reading, and writing, along with demographic and educational background information.

The purpose of this analysis is to identify patterns and factors that may influence student performance and generate meaningful insights from the dataset.

2. Objectives

The main objectives of this project are:

- Analyze student performance data
- Identify trends and relationships between variables
- Compare academic performance across different groups
- Create visualizations to support analysis
- Generate insights using statistical summaries

3. Tools and Technologies

The following tools and technologies were used in this project:

- Python
- pandas
- matplotlib
- Visual Studio Code
- Microsoft Excel

4. Dataset Overview

The dataset contains student exam scores in three subjects:

- Mathematics
- Reading
- Writing

It also includes additional variables such as:

- Gender
- Lunch type
- Parental level of education
- Test preparation course

The dataset was analyzed to evaluate how these factors relate to student academic performance.

5. Data Preparation

Before analysis, the dataset was explored and cleaned. The following steps were performed:

- Loaded the dataset using pandas
- Checked data types and dataset structure
- Verified missing values
- Generated descriptive statistics
- Created a new variable called `average_score`

The average score was calculated using the mean of math, reading, and writing scores.

6. Exploratory Data Analysis

Several analyses and visualizations were created to better understand the data.

Gender and Academic Performance

The analysis showed that female students generally achieved higher average scores in reading and writing subjects, while male students showed relatively stronger performance in mathematics.

Lunch Type and Performance

Students with standard lunch plans achieved higher average scores compared to students receiving free or reduced lunch plans.

Score Distribution

Most students scored within the middle performance range, while fewer students achieved extremely high or low scores.

Top Performing Students

The dataset was sorted to identify the highest-performing students based on average scores across all subjects.

7. Visualizations

The following visualizations were created during the analysis:

- Average Score by Gender (Bar Chart)
- Lunch Type vs Average Score (Bar Chart)
- Distribution of Average Scores (Histogram)

These visualizations helped identify trends and compare performance across different categories.

8. Key Findings

The analysis produced several important insights:

- Female students performed better in reading and writing on average
- Standard lunch plans were associated with higher academic performance
- Mathematics scores showed wider variation compared to other subjects
- Most students achieved moderate average scores

9. Conclusion

This project demonstrated how Python can be used for data cleaning, exploratory data analysis, and visualization. By analyzing student performance data, several meaningful trends and relationships were identified.

The project also highlighted the importance of data visualization in understanding educational performance and communicating findings effectively.

10. Skills Demonstrated

This project demonstrates skills in:

- Data Cleaning
- Exploratory Data Analysis
- Statistical Analysis
- Data Visualization
- Python Programming
- pandas
- matplotlib
- Reporting and Interpretation